

Robert R. Ford

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EDUCATION

PhD, Climate Science, University at Albany, SUNY	Expected April 2027
Dissertation: <i>Mechanisms and impacts of Southern Ocean multidecadal variability</i>	
Advisor: Brian Rose	
BS, Applied Physics & Mathematics, Stockton University	2021

RESEARCH EXPERIENCE

Graduate Research Assistant, University at Albany, SUNY	2021 – Present
Undergraduate Researcher, Stockton University	2019 – 2021

PUBLICATIONS

Submitted

2. **Ford, R. R.**, and B. E. J. Rose: Can internal variability explain observed Antarctic sea ice expansion? *Submitted to The Cryosphere*.
1. Rose, B. E. J., **R. R. Ford**, A. Banihirwe, M. D. Camron, J. Clyne, O. Eroglu, K. FitzGerald, B. Freeman, M. A. Grover, J. Kent, R. May, K. Paul, K. R. Tyle, and A. Zacharias: Pythia Foundations: A community learning resource for Python-based computing in the geosciences. *In revision for Journal of Open Source Education*, <https://jose.theoj.org/papers/6ad4fe1df16342a33f750b43772c1b15>.

Peer-reviewed articles

2. **Ford, R. R.**, and B. E. J. Rose, 2026: A Southern Ocean Multidecadal Oscillator Forced by Deep Convection. *Geophysical Research Letters*, **53**, e2025GL120643, <https://doi.org/10.1029/2025GL120643>.
1. **Ford, R. R.**, B. E. J. Rose, and M. C. Rencurrel, 2025: Transient Climate Sensitivity Shaped by Low Cloud Changes Remotely Driven by Southern Ocean Processes. *Journal of Climate*, **38**, 797–813, <https://doi.org/10.1175/JCLI-D-24-0164.1>.

PRESENTATIONS

6. **Ford, R. R.**, and B. E. J. Rose, 2026: Antarctic sea ice expansion as internal variability: trends and zonal considerations [oral], AGU Annual Meeting.
5. **Ford, R. R.**, and B. E. J. Rose, 2026: Can multidecadal variability of the Southern Ocean explain observed Antarctic sea ice expansion? [poster]. US CLIVAR Broadening Earth System Research Across Scales with High-Resolution Modeling Workshop.
4. **Ford, R.**, and B. E. J. Rose, 2026: Exploring Mechanisms for Southern Ocean Convective Variability in a High-Resolution GCM [poster]. Ocean Sciences Meeting.

3. **Ford, R. R.**, and B. E. J. Rose, 2025: The Role of Low-Frequency Variability and Open-Ocean Polynyas in Southern Ocean SST Trends [oral]. AMS 18th Conference on Polar Meteorology and Oceanography.
2. **Ford, R.**, and B. E. J. Rose, 2023: Transient climate sensitivity shaped by Antarctic sea ice changes: exploring links between ocean heat uptake patterns, sea ice changes, and mid-latitude cloud cover [oral]. AGU Fall Meeting.
1. **Ford, R.**, and J. R. Manson, 2021: Solute transport modeling using a Preissmann scheme [oral]. 6th IAHR Europe Congress.

Coauthored presentations can be found at <https://scholar.google.com/citations?user=PXTHuBAAAAAJ>.

SOFTWARE

Developing [Project Pythia Foundations](#) and [Cookbooks](#) 2021 – Present

Source code can be found at <https://github.com/r-ford> and <https://github.com/ProjectPythia>.

TEACHING

University at Albany, SUNY

Instructor, AATM 103 Introduction to Climate Change Summer 2026

Guest Lecturer, AATM 522 Climate Variability and Predictability Spring 2026

Other

TA, Computational Tools for Climate Science, Climatedmatch Academy Summer 2025

TA, PHYS 2110/2120 Physics for Life Sciences I/II, Stockton University Fall 2019 – Spring 2021

HONORS & AWARDS

Paul Saraduke Jr. Memorial Physics Award, Stockton University 2021

Jason Shulman Award for Excellence in Physics Research, Stockton University 2020

Foundation Scholarship, Stockton University 2020

SERVICE & OUTREACH

Department of Atmospheric & Environmental Sciences, University at Albany, SUNY

Organizer, Climate Group seminar 2024 – 2026

Mentor, Undergraduate–graduate mentorship program 2024 – 2026

Tutor, Volunteer undergraduate tutoring (program lead 2024 – 2025) 2022 – 2025

Professional

Organizer, Project Pythia Cook-off hackathons held at NCAR 2023 – 2026

Contributed presentations on climate modeling to Climatedmatch Academy 2024

TECHNICAL SKILLS

Programming & tools: Python, Git, Bash, NCO, L^AT_EX, HPC environments

Models: CESM1/2

PROFESSIONAL AFFILIATIONS

American Geophysical Union
American Meteorological Society
European Geosciences Union